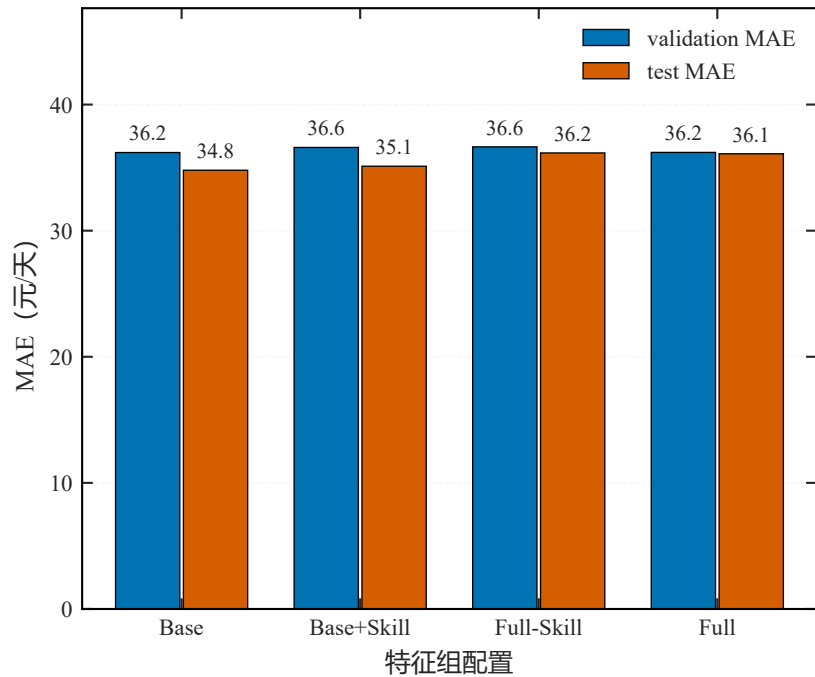
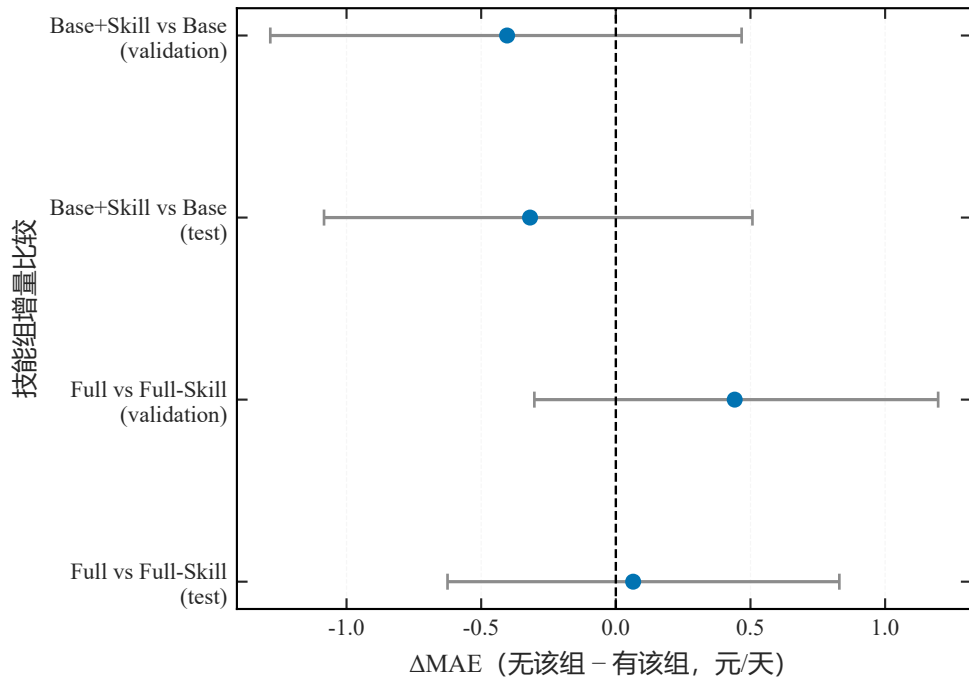




(a) 四配置 validation / test MAE



(b) 技能增量 Δ MAE 与 bootstrap 95% CI

图15 特征组消融 (同一 split / 模型 / 超参数 / 种子, 仅改变特征组): (a) 四配置 validation / test MAE; (b) 技能增量 Δ MAE 与 bootstrap 95% CI (跨 0 即不显著)